

Exam information - MAL1 S26

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General information

Course description:

<https://my.via.dk/media/mitvia/semesteroverblik/kursusbeskrivelser/software-engineering/4.-semester/it-mal1-6-0>

The exam is going to be taking place 8.- 11. June 2026. You can find your timeslot on Wiseflow. If you wish for a different timeslot, you need to find a person to switch with and let me know as soon as possible. Make sure to come 20 minutes before your scheduled time (except for the first person of the day), in case of no-shows. The pre-requisite to attending the exam is $\geq 75\%$ attendance.

The exam is a 20-minute oral examination based on two out six of the course assignments. The list of assignments can be found below. First, you randomly draw which assignments will be the main one and side ones. You will then open your main assignment and present it, after which we will move onto the side assignment. You should not prepare PowerPoint presentations or similar. You are expected to present the underlying theory behind the concepts related to the assignment, the reasoning behind your decisions for code and present the results. You can be asked questions related to both theory, and your practical application. It is therefore very important that you understand your code and how it applies the theory.

You can change your assignments after handing them in. Make sure to bring the latest version with you to the exam. The assessment is based solely on your performance during the exam.

As preparation, I recommend you go through the slides, the relevant chapters in the book, and the additional materials, including the videos. It is crucial that you understand your code and are able to explain what you did and why you did that, rooted in the theory.

Below is the list of the six course assignments and the main topics related to each assignment. Please note that the topics discussed at the exam are **not limited** to only the topics listed under each assignment, and can include any topic discussed in the course. However, these topics are very likely to be discussed, and are a good starting point.

If you collaborated on the assignments with other people, you must make sure that you understand all the code and the decisions behind it, not only the parts that you wrote.

1 Car prices

- Regression algorithms
- Linear regression with linear algebra
- Linear regression with library functions
- Performance metrics for regression
- Regularization with Lasso, Ridge and Elastic Net
- kNN classification

2 Candidates I

- Visualization (box plot)
- Descriptive analysis
- Classification algorithms
- Decision trees
- Random forests
- Gradient-boosted trees

3 Mushroom foraging

- Data preparation
- Logistic regression
- Validation methods
- Confusion matrix
- Performance metrics for classification

4 Detecting exoplanets

- Data preparation
- Data analysis
- Validation methods
- Logistic regression
- SVM
- Confusion matrix
- Performance metrics for classification

5 Sentiment analysis

- Bag of words encoding
- Neural networks
- Hyperparameters for NN
- Activation functions

6 Candidates II

- Dimensionality reduction algorithms
- PCA
- Choice of PC and their meaning
- Clustering algorithms
- K-means
- DBSCAN
- Hierarchical clustering